

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A Scenario-Based Research Assessment

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the
Faculty

Total Marks
Awarded

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of AI Application	Provide the name of the latest AI application selected and the corresponding AI domain.
2	Domain Selection	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics /
3	Introduction	Describe the AI application and provide an overview of its purpose and importance.
4	Problem Identification	Explain the real-world problem or challenge addressed by this AI application.
5	Need for AI Solution	Analyze why AI is required and how it improves traditional approaches.
6	AI Technologies Used	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	Working Mechanism	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	Data Sources and Processing	Describe the type of data used and how the data is processed for intelligent decision-making.
9	Real-World Implementation	Identify organizations, industries, or real-world scenarios where this AI application is currently
10	Impact and Benefits	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	Challenges and Limitations	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	Future Scope and Emerging Trends	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	Conclusion	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	References	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

1.	Idqn tifica tion of La tqs t AI App lica tion	10 %	S q lqc ts a high lyrq lqvan t, qm q rging A l ap plication w ith stro ng jus tification	S q lqc ts a rq cq nt a nd rq lqvan t AI ap p lic atio n.	S q lqc ts asu itab lq ap p lica tion w ith limitqd jus tificatio n .	A pp lica tion iso utda tq d or irrq lqvan t.
2.	Pro blq m Analysis an d D om ain Un dqrsta nd ing	15%	Clqa rly q xp la ins th q rqa l- w o rld p ro blqm , do m a inco ntq xt, a nd nq qd for AI w ith critica l an aly sis .	G o od qx plana tion with ad qq ua tq an aly sis .	Ba sic qx p la na tio nwith lim itq d an alysis .	Pro blqm sta tq m q nt isun clqa r or inco m p lqtq .
3.	AI Tq ch no log iq s an d W o rking M qc ha ni sm	20 %	Accu ratqly qx plains A l tqch niq uq s, alg o rithm s arch itqctu rq , an d wo rkflo w w ith	G o od qx plana tion w ith m ino r o m iss ions.	Co vq rs on ly b as ic A l co nc qp ts.	A l tqc hn o log iqs arq po o rly qx plainq d o r inco rrqct.
4.	Rq sq a rch Q ua lity an d Litq ra turq Rq viqw	10 %	U sq s rq cqn t, crq diblq s o urcq sw ith pro pq r cita tio ns an d qx cq llqnt sy nthq sis .	G o od qua lity rq fq rq nc qs with ap propria tq citations .	Lim itqd rq fq rq nc qs w ith b as ic d iscu ss ion.	F qw o r un rq liablq rq fq rq nc qs ;po o r c ita tion.
5.	Rq al-W orld Im plq m q nta tion an d Ca s q A na lysis	10 %	Th oro ug hly an alyzqs indus trial im plq m qn tatio n w ith a pprop ria tqqx am plq s.	G o od im plq m qn t at ion an aly sis .	Ba sic im plq m qn tation d iscu ss ion.	M inim al o r no rqa l-w o rld an aly si s .
6.	Im p ac t, Cha llq ng q s, an d Futu rq S co pq	15%	Critic a lly qv a luatq s bq n qfitslim ita tions , qthical is s uq s , an d futu rq trq nd s .	G o od qv a luation w ith m ino r g ap s .	Ba s ic d iscu ss ion o fim p a ct a nd futu rq sc op q .	Lim itq d qv a luation o rm iss in g s q ctions .
7.	Origina lity an d Critica l Th inkin g	10 %	D q m o ns tratqs indq pq ndq nt rq s q arch , inn o va tion, an d critica l q va luation.	G o od an aly ticalthin king .	S o m q o rigina lity w ith limitqd an alysis .	M ostly dq s criptivq o r co piq d co ntq nt.
8.	O rg an iz atio n, Prq s qn tatio n, an d Do cu m q nta tion	10 %	E xcq p tion a lly o rgan izqd , profq ss iona l prqs q nta tion, qrror frqq , an d properly rq fq rq nc qd .	W qll o rgan izqd w ith m ino r iss uq s.	A dq quatq ly o rgan izqd w ith so m q q rrors.	Po or org an izatio nand do c um q ntation.

SiqmQns Industrial Copilot - AIApplication in Smart Manufacturing
Systqm Architqcturq of SiqmQns IndustrialCopilot

Thq SiqmQns Industrial Copilotopqratqs using a multi-layer
architqcturq that connQcts factory QquipmQnt, QnginQQring
softwarq, cloud sqrvicqs, and artificial intqlligQncq modQls. This

architecture enables seamless communication between machines, engineering, and AI systems, allowing manufacturing processes to become more intelligent and efficient.

The first layer consists of physical industrial equipment such as Programmable Logic Controllers (PLCs), industrial robots, conveyor systems, sensors, actuators, motors, drives, and Human Machine Interfaces (HMIs). These devices continuously collect operational data during manufacturing processes. They monitor machine conditions, execute automation commands, and communicate with higher-level systems.

The second layer is the Industrial Internet of Things (IIoT) layer. This layer gathers real-time information from factory equipment, including machine status, production counts, temperature, pressure, vibration levels, energy consumption, alarm messages, and maintenance records. The collected data forms the foundation for AI-based analysis and decision-making.

The third layer is Siemens' engineering environment, particularly the Total Integrated Automation (TIA) Portal. Engineers use TIA Portal to configure PLCs, develop automation programs, simulate machine operations, and monitor industrial processes. Siemens IndustrialCopilot integrates directly with this environment, enabling AI-generated PLC code to be inserted into engineering projects quickly and accurately.

The fourth layer is Microsoft Azure Cloud, where powerful AI models perform natural language understanding, PLC code generation, engineering analysis, and intelligent recommendations. Azure provides scalable cloud computing resources capable of processing large industrial datasets while maintaining enterprise-grade security and reliability.

Finally, engineers interact with the Industrial Copilot through a conversational interface. Instead of writing complex PLC programs manually, engineers simply describe their automation requirements using natural language. The AI interprets the request, generates suitable automation code, explains the logic, and recommends improvements based on industrial best practices. This layer

architecture enables secure communication, efficient engineering collaboration, and intelligent automation while reducing development time and improving manufacturing productivity.

Features of Siemens Industrial Copilot

Siemens Industrial Copilot provides numerous intelligent features that simplify engineering activities and improve manufacturing efficiency.

One of its most valuable features is automatic PLC code generation. Engineers can describe machine operations using natural language, and the AI generates Ladder Logic, Structured Text (ST), or Function Block Diagram (FBD) code that follows Siemens programming standards.

The Industrial Copilot also generates engineering documentation automatically. Instead of manually preparing technical documents, the AI creates machine descriptions, operating procedures, maintenance instructions, and project documentation, saving engineers considerable time.

Another important capability is code explanation. The AI explains every section of the generated PLC program, helping engineers understand the control logic and making it easier for beginners to learn industrial automation programming.

Fault diagnosis is another powerful feature. The system analyzes alarm messages, sensor readings, PLC programs, and machine behavior to determine possible causes of equipment failures. It then recommends troubleshooting procedures to minimize machine downtime.

Predictive maintenance is supported through continuous monitoring of industrial equipment. By analyzing equipment conditions and historical maintenance records, the AI identifies early warning signs of potential failures before breakdowns occur, allowing maintenance teams to schedule repairs proactively.

The Industrial Copilot also acts as a digital knowledge assistant. It provides instant answers to engineering questions based on technical manuals, engineering standards, previous automation

projects, and Siemens documentation. This preserves organizational knowledge and assists newly hired engineers.

As the system processes additional industrial data, its recommendations become increasingly accurate, allowing continuous improvement in engineering performance and decision-making.

Comparison Between Traditional Automation and AI-Based Automation

Traditional industrial automation relies heavily on human expertise. Engineers manually develop PLC programs, design control systems, test automation logic, diagnose machine failures, and prepare technical documentation. Although these methods have been used successfully for many years, they require significant engineering effort, specialized knowledge, and considerable time.

AI-based automation transforms this process by introducing intelligent engineering assistance. Engineers no longer need to write every line of PLC code manually. Instead, they simply describe the required machine operation in natural language. Traditional troubleshooting often requires engineers to examine electrical drawings, PLC programs, sensor values, alarm logs, and maintenance records before identifying equipment faults. This process can consume several hours or even days. Siemens Industrial Copilot analyzes machine data almost instantly and provides likely causes of failures together with recommended corrective actions.

Documentation is another area where AI provides substantial improvements. Manual documentation often requires additional engineering effort and increases the possibility of inconsistencies. The Industrial Copilot automatically produces standardized documentation, improving accuracy and reducing workload.

Overall, AI-assisted automation reduces engineering effort,

accelerates project completion, improves code quality, minimizes downtime, and increases manufacturing productivity.

Case Study - Conveyor Belt Automation

Consider a manufacturing company operating an automated conveyor belt system that transports packaged products between different production stations.

Traditionally, an automation engineer would spend several days designing PLC logic, configuring sensors, implementing emergency stop functions, testing automation sequences, and preparing technical documentation.

Using Siemens Industrial Copilot, the engineer simply enters a request such as:

"Create a PLC program for a conveyor belt with start and stop buttons, emergency stop, overload protection, object detection sensor, and automatic restart after power recovery."

The AI analyzes the engineering requirements and automatically generates Ladder Logic, Structured Text code, input-output mapping, timer configurations, safety recommendations, and technical documentation.

The engineer reviews the generated program, performs simulation within Siemens TIA Portal, verifies safety requirements, and downloads the PLC program to the production system. During normal operation, the Industrial Copilot continuously monitors conveyor performance. If the conveyor unexpectedly stops because of an overload or sensor malfunction, the AI analyzes alarm messages and recommends troubleshooting procedures. Maintenance engineers can quickly identify the root cause and restart production with minimal downtime.

This example demonstrates how AI significantly reduces engineering time while improving production reliability and maintenance efficiency.

Cybersecurity in Siemens Industrial Copilot

As industrial automation systems become increasingly connected through cloud computing and Industrial IoT technologies, cybersecurity becomes a critical requirement. Siemens Industrial Copilot incorporates several security mechanisms to protect engineering projects and industrial data. User authentication ensures that only authorized personnel can access automation systems, while multi-factor authentication provides an additional layer of protection.

Communication between factory equipment and Microsoft Azure cloud services is encrypted using secure communication protocols. Encryption protects industrial information from unauthorized access during transmission.

Role-based access control allows organizations to assign different permissions to engineers, operators, supervisors, and administrators. This ensures that users can only access the information necessary for their responsibilities.

The platform also receives regular security updates that protect against newly discovered cyber threats. Compliance with international industrial cybersecurity standards helps organizations maintain secure and reliable manufacturing operations.

Revolutionizing Engineering with Generative AI

The Siemens Industrial Copilot is the first generative AI-powered assistant designed explicitly for industrial environments. Unlike traditional tools, this copilot goes beyond automation to actively support engineers in code creation, configuration, and process optimization. By integrating natural language processing and machine learning, Siemens enables engineers to leverage AI without specialized expertise, significantly speeding up project timelines and reducing repetitive tasks.

As Rainald Brähm, CEO of Factory Automation at Siemens, emphasized, “The Siemens Industrial Copilot is a supercharger for industrial automation, accelerating innovation, productivity, and competitiveness.” The copilot’s capabilities align with Gartner’s projections that, by 2028, 75% of developers will regularly use AI to assist with code creation.

Practical Applications at Thyssenkrupp and Siemens Electronics Factory

Thyssenkrupp Automation Engineering has integrated the Industrial Copilot for Engineering to enhance the efficiency of machines used in electric vehicle battery quality inspections. This AI assistant generates structured control language (SCL) code for programmable logic controllers, automatically integrates it within the TIA Portal, and generates machine visualizations. The result is a faster, more intuitive design and engineering process, which reduces time-to-market and minimizes operational errors.

Simultaneously, Siemens Electronics Factory in Erlangen implemented the copilot across its soldering machines. The Industrial Copilot supports operators by translating machine error codes into natural language and suggesting solutions based on operational history and technical documentation. This feature dramatically cuts machine downtime, helping operators resolve issues quickly and reducing production bottlenecks.

Future of Industrial Copilot: Multimodal and Agent-Based Capabilities

Siemens plans to evolve the Industrial Copilot by incorporating multimodal capabilities and agent-based automation. The multimodal function will enable the copilot to process and analyze images, further enhancing productivity in maintenance and inspection tasks. The agent-based approach will autonomously manage complex tasks by breaking them down into actionable subtasks, offering real-time insights, and automating decision-making on the shop floor.

Embracing Data Sovereignty with On-Premises Solutions To ensure data security, Siemens offers an on-premises configuration for the Industrial Copilot, utilizing the Simatic Industrial PC (IPC 1047E) powered by NVIDIA AI Enterprise

software. This approach allows data processing directly on the shop floor, eliminating the need for an internet connection and ensuring that sensitive data remains within the facility. This on-premises setup meets stringent data sovereignty requirements, providing peace of mind for customers in regulated

industriQs.

A New Era for AI in Industrial Automation

With these advancements, Siemens solidifies its leadership in industrial automation by delivering an AI-driven copilot that combines scalability, data security, and unmatched usability. The Industrial Copilot offers manufacturing leaders a strategic advantage: faster engineering, streamlined operations, and reduced downtime. As industries increasingly adopt generative AI, Siemens' commitment to innovation ensures that manufacturing remains competitive, efficient, and future-ready.



